

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Mathematics

2. Subject Name: Numerical Methods Laboratory **L-T-P:** 0-0-3 **Credits:** 2

3. Pre-requisites: Applied Computational Methods, Any advanced level programming language

4. Syllabus and reference books:

Numerical solutions of non-linear equations in one-variable using simple enclosure methods and fixed-point iteration methods; Approximations using linear interpolation, cubic spline, least square polynomials, Numerical solutions of ODEs: initial value problems and boundary value problems; Numerical solutions for parabolic and elliptic PDEs using finite difference and iterative methods; Numerical solutions of conservation laws using finite difference, finite volume, slope limiters, and iterative methods.

Reference Books:

1. Joe D. Hoffman, Numerical Methods for Engineers and Scientists, CRC Press.
2. E. F. Toro Riemann solvers and numerical methods for fluid dynamics: A practical introduction, Springer-Verlag, Berlin, Heidelberg, third ed. 2009.
3. J. S. Hesthaven: Numerical methods for conservation laws, vol. 18 of Computational Science & Engineering, Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 2018. From analysis to algorithms.

1. Lecture-wise break-up:

Sl. No.	Topic	No. of Labs
1.	Numerical solutions of non-linear equations in one-variable	2
2.	Polynomial Approximations	2
3.	Numerical solutions of ODEs	2
4.	Numerical solutions for parabolic and elliptic PDEs	2
5.	Numerical solutions of conservation laws	4
Total number of hours		36